Tertiary Infotech Academy Pte Ltd

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FACILITATOR GUIDE

For

Agentic AI Applications with Claude Code

TGS Ref No: TGS-2025052468

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	22 Jan 2025	First version	Tertiary Infotech Academy
2.0	17 Oct 2025	Update company name	Tertiary Infotech Academy
3.0	12 October 2026	Full revision for the current course title Agentic AI Applications with Claude Code, rebuilt to the WSQ house standard (cover page, version control record, page-numbered table of contents, footer). TSC, K&A, LUs, LOs, methods, ratios, oral clarification (up to 10 min) and venue as in the approved CP; the delivered Claude Code courseware (deck v17, LP v1.9, LG v1.8, labs) is mapped to each LU. Lesson plan follows LP v1.9 re-timed to the CP-approved 7 training hours and 1 assessment hour (LU1 2 h, LU2 2 h, LU3 3 h; WA-SAQ 30 min, PP 30 min). Assessment appendix added.	Dr. Alfred Ang
4.0	31 October 2026	Aligned with deck v23 (83 slides), LP v2.2 and LG v2.1: delivery map and lesson-plan slide ranges recomputed; Topic 1.1 adds what makes an AI agent (model + harness, seven parts); Topics 3.1/3.2 add SKILL.md and sub agent front matter.	Tertiary Infotech Academy Pte Ltd
5.0	7 November 2026	Aligned with deck v25 (91 slides), LP v2.3 and LG v2.2: one topic, one activity, one lab and one PP task. Setup now covers four surfaces, including the cloud at claude.ai/code, with slides for saved Desktop environments and cloud sessions. Agent Skills moved to Topic 2 (Commands, Tools and Skills); Topic 3 is now Agents, Hooks and Security, with AI security taught as five principles. Activities renumbered 1, 2A–2D and 3A–3C; WhatsApp widget activity removed; learners build their own hooks. LU1 and LU2 timing unchanged; the LU3 block re-timed within the same 3 hours. Assessment appendix removed: assessment content now lives only in the Assessment Plan, so this guide can be published. Labs repository link corrected.	Tertiary Infotech Academy Pte Ltd

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TABLE OF CONTENTS

1. Course Overview....4
1.1 TSC Knowledge and Abilities....4
1.2 Abbreviations....4
2. Learning Units, Learning Outcomes and K&A Mapping....5
2.1 Delivery Map — Learning Units to the Current Courseware....6
3. Facilitator Preparation....7
3.1 Resources Required....7
3.2 Before the Class....7
3.3 Facilitator Roles and Responsibilities....7
4. Instructional Methods....8
5. Lesson Plan and Facilitator Instructions....9
Day 1 — Build, deploy and extend an AI-built website with Claude Code, then assessment....9
5.1 Facilitation Notes....11
6. Assessment Overview....11
6.1 Structure and Duration of Training and Assessment....11
6.2 Summary of Assessment Requirements....11
6.3 Assessment Methods, Duration and Ratio....11
6.4 Conducting the Assessment....12
7. Reference Documents....12

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1. Course Overview

Course Title	Agentic AI Applications with Claude Code
TGS Ref No	TGS-2025052468
Skills Framework	Engineering Services
TSC Title	Artificial Intelligence Application
TSC Code	EGS-TEM-3026-1.1
TSC Category	Technology Road Mapping
TSC Description	Deploy Artificial Intelligence (AI) workflows for enhancing the efficiency of engineering and maintenance processes
TSC Proficiency Level	3
Course Duration	1 day · 8 hours (7 hours training: 3.5 hrs classroom facilitation + 3.5 hrs practical; 1 hour assessment)
Daily Timing	9:00 am – 5:45 pm (45-minute lunch not counted; tea breaks within training time)
Mode of Delivery	Classroom (physical), instructor-led: lecture, didactic questioning, demonstration and hands-on practice building, deploying and extending a website with Claude Code
Class Size / Ratio	Maximum 20 learners per trainer (CP trainer-to-learner ratio 1:3 min / 1:20 max)
Assessment	Written Assessment – Short-Answer Questions (WA-SAQ) — 30 min, individual, summative, open book Practical Performance (PP) — 30 min, individual, summative, open book

1.1 TSC Knowledge and Abilities

Code	Statement	Assessed by
K1	Background and basics of AI	WA-SAQ
K2	Fundamental concepts and methods of statistics and programming for data science	WA-SAQ
K3	Mathematics and computing theories	WA-SAQ
K4	Machine learning techniques and applications	WA-SAQ
K5	Principles of data management	WA-SAQ
K6	Software development methodologies	WA-SAQ
K7	Types of AI applications	WA-SAQ
K8	AI implementation procedures	WA-SAQ
A1	Implement AI applications in collaboration with technology service providers	PP
A2	Deploy AI workflows according to plan	PP
A3	Identify and report any issues with the AI applications and data collected	PP

1.2 Abbreviations

Abbreviation	Meaning
LG / FG / AP / LP	Learner Guide / Facilitator Guide / Assessment Plan / Lesson Plan
CP	Course Proposal (TPGateway TPG-2025086933)
TSC / K / A	Technical Skills and Competency / Knowledge statement / Ability statement
LU / LO	Learning Unit / Learning Outcome
WA-SAQ / PP	Written Assessment – Short-Answer Questions / Practical Performance
C / NYC	Competent / Not Yet Competent
MCP	Model Context Protocol

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2. Learning Units, Learning Outcomes and K&A Mapping

The Learning Units (LUs), Learning Outcomes (LOs) and Knowledge & Abilities (K&A) below are as approved in the Course Proposal; the topic lists reflect the current courseware. Section 2.1 shows where each LU is delivered in the current deck and activities.

Learning Unit	Topics (K&A)	Learning Outcome and K&A Statements
LU1: Amazon Bedrock and Foundation Models	Approved CP content — Topic 1: Amazon Bedrock and Foundation Models (K1, K2, K3, A1) • Introduction to Amazon Bedrock • Foundation Models • Model Families and Capabilities • Fine-Tuning and Customization • Application Components • Core Building Blocks of a Generative AI Application • Architecture Fundamentals Delivered in courseware v25 as Topic 1: Claude Code Fundamentals (K1, K2, K3, A1) • Claude product family; evolution of AI engineering (prompt → context → harness → harness systems) • Harness engineering; what makes an AI agent (model + harness + tokens; seven parts); what Claude Code is and can do; setup (terminal, VS Code, desktop, cloud) and pricing • The agentic loop and built-in tools; the 7 steps to build an app with Claude Code • Permission modes and bypass permissions • Context engineering, the context window, memory files (CLAUDE.md, auto memory), sessions, scheduled tasks and /goal	LO1: Implement AI applications using AWS Bedrock and Foundation Models K1: Background and basics of AI K2: Fundamental concepts and methods of statistics and programming for data science K3: Mathematics and computing theories A1: Implement AI applications in collaboration with technology service providers
LU2: AI Workflow Using LangChain	Approved CP content — Topic 2: AI Workflow Using LangChain (K4, K6, K8, A2) • LangChain Basics • Building Pipelines with LangChain • Best Practices and Optimization • Environment Setup • Building a LangChain Pipeline • Testing and Validation Delivered in courseware v25 as Activity 1 (7-step build workflow) and Topic 2.1: Custom Commands (K4, K6, K8, A2) • Activity 1a/1b — build a software-development website with Claude Code following the 7 steps: goal & features, plan mode, execute, /init context, commit to GitHub, GitHub Pages deployment, CI/CD with GitHub Actions • The .claude directory; user level vs project level • Custom slash commands and $ARGUMENTS; Activity 2A — /gitpush command (push, README, Pages, About, security scan)	LO2: Deploy AI workflows leveraging LangChain and machine learning techniques. K4: Machine learning techniques and applications K6: Software development methodologies K8: AI implementation procedures A2: Deploy AI workflows according to plan
LU3: Retrieval-	Approved CP content — Topic 3: Retrieval-	LO3: Identify and report issues in

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Learning Unit	Topics (K&A)	Learning Outcome and K&A Statements
Augmented Generation (RAG) and Agents	Augmented Generation and Agents (K5, K7, A3) • Explore Generative AI Use Cases Using Amazon Bedrock • Using Knowledge Bases • Build and Evaluate RAG Applications Using Amazon Bedrock Knowledge Bases • RAG Concepts in Practice • Implementation Steps • Evaluation and Iteration • Using Agents • Agent Overview • Designing Agent-Based Workflows • Governance and Control Delivered in courseware v25 as Topic 2.2 (MCP tools), Topic 2.3 (Agent skills) and Topic 3: Agents, Hooks and Security (K5, K7, A3) • MCP architecture, connecting servers and scopes, Playwright MCP; Activity 2B/2C — screenshot the site and test the enquiry form end to end • Agent skills, inside a SKILL.md (name and description front matter), SKILL.md vs CLAUDE.md; Activity 2D — install and run frontend-design and seo-audit skills and redesign the site • Sub agents (front matter: name, description, tools, skills, model; why use them; sub agents vs agent teams); Activity 3A — security and UI/UX sub agents that report issues and fixes • Hooks and lifecycle events; PreToolUse and PostToolUse use cases; Activity 3B — build a pop-up hook and a voice hook • AI security — data control, permission control, human in the loop, traceability, right-sized models; Activity 3C — test, secure and redeploy; key concepts summary — what triggers what	Retrieval-Augmented Generation (RAG) and AI Agents applications. K5: Principles of data management K7: Types of AI applications A3: Identify and report any issues with the AI applications and data collected

2.1 Delivery Map — Learning Units to the Current Courseware

Learning Unit	Delivered in (current deck / activities)	K&A
LU1: Amazon Bedrock and Foundation Models	Topic 1 — Claude Code Fundamentals, including what makes an AI agent (slides 14–40) and Topic 1 context, memory and sessions (slides 42–49); Activity 1A — Goal, Plan & Build (slides 30–36, 41).	K1, K2, K3, A1
LU2: AI Workflow Using LangChain	Activity 1A/1B — Build & Deploy a Website (slides 41, 50–53; labs/lab1-seven-steps); Topic 2.1 — .claude directory and Custom Commands with Activity 2A (slides 54–58; labs/lab2-commands-tools-skills Part A).	K4, K6, K8, A2
LU3: Retrieval-Augmented Generation (RAG) and Agents	Topic 2.2 — MCP Tools with Activities 2B and 2C (slides 59–63); Topic 2.3 — Agent Skills with Activity 2D (slides 64–67; labs/lab2-commands-tools-skills);	K5, K7, A3

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Learning Unit	Delivered in (current deck / activities)	K&A
	Topic 3 — Agents, Hooks and Security with Activities 3A–3C (slides 68–86; labs/lab3-agents-hooks-security); key-concepts summary (slide 87).	

3. Facilitator Preparation

3.1 Resources Required

- Trainer slide deck v25 (PPT, 91 slides) and learner slides (PDF), Learner Guide v2.2 and Lesson Plan v2.3 — download from the LMS at https://lms-tms.tertiaryinfotech.com/
- Lab repository https://github.com/tertiarycourses/TGS-2025052468-Agentic-AI-Applications-with-Claude-Code/tree/main/labs (lab1-seven-steps, lab2-commands-tools-skills, lab3-agents-hooks-security)
- Learner laptops (macOS, Linux or Windows) with internet access, Claude Code installed and signed in (any paid Claude plan or Console account), Node.js (for npx) and Git
- A GitHub account per learner (repository, GitHub Pages and GitHub Actions)
- 75" TV / projector for demonstrations, whiteboard and markers, high-speed Wi-Fi; spare laptops for learners without one
- TRAQOM digital attendance QR codes (AM, PM and Assessment); WA-SAQ and PP papers on the LMS

3.2 Before the Class

- Download the current deck (v25), Learner Guide, Lesson Plan and the lab repository; confirm the slide ranges against this guide.
- Check the venue network allows claude.ai, github.com, npm and the Playwright MCP download; test the Claude Code install commands on a venue machine.
- Remind learners before class to bring a laptop with Claude Code signed in, a GitHub account and Node.js installed.
- Prepare the TRAQOM QR codes for the AM, PM and Assessment digital attendance.
- Pre-load the WA-SAQ and PP papers on the LMS; keep the Assessment Plan, which holds the marking guide, with the assessor only.

3.3 Facilitator Roles and Responsibilities

- Deliver every topic and activity in the sequence of the Lesson Plan, using the slide ranges shown.
- Take the mandatory AM, PM and Assessment digital attendance (TRAQOM QR) at the points shown in the lesson plan.
- Check understanding through questioning; coach each learner during the hands-on activities.
- Conduct the Briefing for Assessment, then administer the assessment exactly as specified in the Assessment Plan.
- Record outcomes in the Assessment Summary Record, give feedback, and upload the assessment evidence to the LMS.
- Remind learners to complete the TRAQOM survey and course feedback before they leave.

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4. Instructional Methods

Instructional Method	How it is used in this course	Trainer : Learner
Lecture	Concept briefings for each topic using the slide deck — harness systems, the agentic loop, context and memory, MCP, skills, sub agents, hooks and AI security.	1:3 (min) / 1:20 (max)
Didactic Questioning	Pre-planned questions after each segment to check learners can explain when and why each mechanism is used (e.g. CLAUDE.md vs SKILL.md, /compact vs /clear).	1:3 (min) / 1:20 (max)
Demonstration	The trainer runs each Claude Code step live (plan mode, /init, MCP install, /agents, hooks) before learners attempt it.	1:3 (min) / 1:20 (max)
Practice	Learners build, deploy, test and extend their own website through Activities 1–3, each verified against its checkpoint.	1:3 (min) / 1:20 (max)

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5. Lesson Plan and Facilitator Instructions

Sequence, topics, activities and slide content follow Lesson Plan v2.3; slide numbers refer to the current 91-slide trainer deck v25. LP v2.3 runs 9:00 am – 6:00 pm; this guide re-times the day to the CP-approved 7 training hours and 1 assessment hour (9:00 am – 5:45 pm) and to the CP time per Learning Unit (LU1 2 h, LU2 2 h, LU3 3 h, inclusive of tea breaks, attendance and briefings falling within each LU). Activity 1 is extended to 90 min within LU2; the welcome and the Topic 2 and Topic 3 blocks are shortened; the assessment block moves to 4:45–5:45 pm.

Day 1 — Build, deploy and extend an AI-built website with Claude Code, then assessment

Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
9:00–9:20	20 min	Welcome, digital attendance (AM), trainer and learner introductions, ground rules, course outline, lab materials, lesson plan, briefing for assessment and the LMS.	Lecture	Slides 1–13, QR
9:20–10:15	55 min	LU1 · Topic 1.1 Claude Code fundamentals (K1, K2, K3, A1) — Claude product family; evolution of AI engineering; harness engineering; what makes an AI agent (model + harness + tokens; seven parts); what Claude Code is and can do; setup in terminal, VS Code, desktop and the cloud (claude.ai/code); pricing; the agentic loop and built-in tools; the 7 steps to build an app; permission modes and bypass permissions. Activity 1a: goal, plan and build (steps 1–3). Facilitator: ask learners to name what the harness adds around the model.	Lecture, Didactic Questioning, Demonstration, Practice	Slides 14–41
10:15–10:25	10 min	Tea break	—	—
10:25–11:00	35 min	LU1 · Topic 1.2 Context, memory and sessions — context engineering (CLAUDE.md, @ mentions, /compact, /clear); the context window and /context; memory files (CLAUDE.md via /init, auto memory and /memory); managing sessions; scheduled tasks (Routines); /goal. Facilitator: check learners can say when to /compact and when to /clear.	Lecture, Didactic Questioning, Demonstration	Slides 42–49
11:00–12:30	90 min	LU2 · Activity 1 — Build & Deploy a Website (K4, K6, K8, A2): complete the 7-step workflow — /init context, commit to GitHub, deploy on GitHub Pages via GitHub Actions, add a README, push a change to confirm CI/CD redeploys the site (labs/lab1-seven-steps). Facilitator: check every learner has a live GitHub Pages URL with the hero, three testimonial cards and the enquiry form.	Demonstration, Practice	Slides 41, 50–53, Lab 1
12:30–13:15	45 min	Lunch break	—	—
13:15–13:45	30 min	LU2 · Topic 2.1 Custom Commands — PM digital attendance; the .claude directory; user level vs project level; custom slash commands and $ARGUMENTS. Activity 2A: create and run a /gitpush command (push, README, Pages, About link, security scan).	Lecture, Demonstration, Practice	Slides 54–58, QR, Lab 2
13:45–14:20	35 min	LU3 · Topic 2.2 MCP Tools (K5, K7, A3) — what MCP is (server, client, transport, tools); connecting servers and scopes; Playwright MCP. Activity 2B: install Playwright MCP, screenshot the live site into the README.	Lecture, Demonstration, Practice	Slides 59–63, Lab 2

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Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
		Activity 2C: test the enquiry form end to end and confirm enquiries reach the inbox. Facilitator: have learners report any defect the Playwright test finds and how it was fixed.		
14:20–14:55	35 min	LU3 · Topic 2.3 Agent Skills — SKILL.md and its front matter (name, description), project skills, installing skills; SKILL.md vs CLAUDE.md. Activity 2D: install and run the frontend-design and seo-audit skills and redesign the site (labs/lab2-commands-tools-skills). Facilitator: tell learners to keep a screenshot of the redesigned site; ask which words in their prompt made the skill fire.	Lecture, Demonstration, Practice	Slides 64–67, Lab 2
14:55–15:05	10 min	Tea break	—	—
15:05–15:30	25 min	LU3 · Topic 3.1 Sub Agents — creating sub agents with /agents and their front matter (name, description, tools, skills, model); why use sub agents; sub agents vs agent teams. Activity 3A: security sub agent and lead-magnet UI/UX sub agent — review the issues each reports and apply the fixes.	Lecture, Demonstration, Practice	Slides 68–73, Lab 3
15:30–16:05	35 min	LU3 · Topic 3.2 Hooks — hook lifecycle events and automating with hooks; PreToolUse use cases (validate or block) and PostToolUse use cases (act on the outcome, such as processing a refund); demo the 10-second pop-up and the voice message on chatbot click. Activity 3B: learners build their own hooks. Facilitator: ask each learner which event fires their hook and what it does; tell them to keep a screenshot of a hook they built.	Lecture, Demonstration, Practice	Slides 74–79, Lab 3
16:05–16:35	30 min	LU3 · Topic 3.3 AI Security — data control (classification), permission control (authorisation: read, write, edit, send), human in the loop (gating and escalation), traceability (logging and a data-leak case), right-sized models. Activity 3C: test, secure and redeploy; learners add deny rules and an audit log. Facilitator: ask learners which tier their project data sits in and who should approve a push.	Lecture, Demonstration, Practice	Slides 80–86, Lab 3
16:35–16:45	10 min	LU3 · Key concepts summary (what triggers what), Q&A, assessment flow, TRAQOM survey and Assessment digital attendance.	Didactic Questioning	Slides 87–91, QR
16:45–17:15	30 min	Written Assessment — Short-Answer Questions (WA-SAQ): 8 questions (K1–K8), open book, on the LMS.	Assessment	WA-SAQ paper
17:15–17:45	30 min	Practical Performance (PP): 3 screenshot-evidenced tasks on the candidate's own website, open book, on the LMS: one task per topic and activity. Sign the	Assessment	PP paper

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Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
		Assessment Summary Record.		

Total: 8 hours (7 hrs training + 1 hr assessment), 9:00 am – 5:45 pm, excluding the 45-minute lunch.

5.1 Facilitation Notes

- Learners work on their own website throughout: the Activity 1 site is extended in every later activity and is the evidence for the PP.
- Bypass permissions is used only in the learner's own sandboxed project; never in a folder holding personal data or credentials.
- Never commit API keys, tokens or personal data; the /gitpush command and the security sub agent must report a clean scan before each push.
- Learners who fall behind should clone the matching reference site or lab step from the lab repository so they can continue with the next activity.
- One topic, one activity, one PP task: remind learners to keep a screenshot at the end of Activity 1 (live site), Activity 2D (redesign) and Activity 3B (a hook they built).
- A minimum of 75% attendance is required for assessment and funding eligibility.

6. Assessment Overview

6.1 Structure and Duration of Training and Assessment

Component	Duration
Training contact time (inclusive of tea breaks): classroom facilitation 3.5 hrs + practical 3.5 hrs	7 hrs
Written Assessment – Short-Answer Questions (WA-SAQ)	30 min
Practical Performance (PP)	30 min
Total course duration	8 hrs

6.2 Summary of Assessment Requirements

Learning Outcome	Assessment Method(s)
LO1: Implement AI applications using AWS Bedrock and Foundation Models	WA-SAQ, PP
LO2: Deploy AI workflows leveraging LangChain and machine learning techniques.	WA-SAQ, PP
LO3: Identify and report issues in Retrieval-Augmented Generation (RAG) and AI Agents applications.	WA-SAQ, PP

6.3 Assessment Methods, Duration and Ratio

Method	K&A covered	Questions / Tasks	Duration	Assessor : Candidate
Written Assessment – Short-Answer Questions (WA-SAQ) Individual, summative, open book	K1, K2, K3, K4, K5, K6, K7, K8	Question 1 — K1 Question 2 — K2 Question 3 — K3 Question 4 — K4 Question 5 — K5 Question 6 — K6	30 min	1:3 (min) / 1:20 (max)

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Method	K&A covered	Questions / Tasks	Duration	Assessor : Candidate
		Question 7 — K7 Question 8 — K8		
Practical Performance (PP) Individual, summative, open book	A1, A2, A3	Task 1 — A1 Task 2 — A2 Task 3 — A3	30 min	1:3 (min) / 1:20 (max)

6.4 Conducting the Assessment

1. Deliver the Briefing for Assessment: purpose, methods, duration, open-book rules, the C/NYC grading and the appeal process.
2. Take the Assessment digital attendance and verify each candidate's identity (name and last 3 digits + alphabet of NRIC/FIN).
3. Issue the question papers only at the start of each assessment; keep the Assessment Plan with the assessor.
4. Invigilate; candidates work individually. Answer clarification questions about the task, never about the answer.
5. Mark each response against the rubric in the Assessment Plan; use oral clarification (1:1, up to 10 minutes) only to close a minor gap.
6. Record C/NYC per LO in the Assessment Summary Record, give feedback, obtain both signatures, and upload the scripts and records to the LMS.
7. A candidate assessed NYC is offered re-assessment in line with the Assessment Plan; explain the appeal procedure if the candidate disagrees.

7. Reference Documents

Document	Purpose
Slides v25, LP v2.3, LG v2.2, lab repository	Teaching material, sequence, activities and labs
Assessment Plan; WA-SAQ paper v4 and PP paper v5	Assessment conditions, records and marking guide; the instruments
Course Proposal TPG-2025086933	Approved LUs, LOs, K&A, methods and durations

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